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20IT3T01 : DISCRETE MATHEMATICS				

COURSE OBJECTIVES:

- Familiarise closed form solution of linear recurrence relations by various methods.
- To introduce basics of set theory and its applications
- Bring awareness of basic concepts of graphs and its applications.
- To teach the topics on Trees, spanning trees, minimal spanning trees and justification of Kruskal's algorithm.

COURSE OUTCOMES:

Upon successful completion of this course the student should be able to

CO1: Identify programming errors efficiently through enhanced logical capabilities (k_3)

CO2: Find a general solution of recurrence equation (k_3)

CO3: Learn set theory, graph of the relations which are used in data structures (k_3)

CO4: Explain the concepts in graph theory (k_3)

CO5: Apply graph theory concepts in core subjects such as data structures and network theory effectively. (k_3)

UNIT 1: MATHEMATICAL LOGIC

Connectives, negation, conjunction, disjunction, statement formula and Truth Tables, conditional and bi-conditional, well formed formulae, tautologies, equivalence of formulae, duality, tautological implications, functionally complete set of connectives, other connectives, principal disjunctive and conjunctive normal forms, inference calculus, rules of inference, consistency of premises, indirect method of proof, Theory of inference for the statement calculus, validity using Truth tables.

Learning Outcomes:

After completion of this unit, student will be able to

- find equivalence formulas, implementation of logic for mathematical proofs (k_1)
- apply inference theory to verify the consistence of data (k_3)

UNIT 2: RECURRENCE RELATIONS

Generating Function of Sequences, Calculating Coefficient of generating functions, Recurrence relations, solving recurrence relation by substitution and Generating functions, the method of Characteristic roots, Solution of In homogeneous Recurrence Relation.

Learning Outcomes:

After completion of this unit student will be able to

- formulate recurrence relations of the sequences (k_3)
- solve homogeneous linear recurrence relations (k_3)
- evaluate complementary function and particular integral for non homogeneous linear recurrence relations (k_3)
- apply substitution method to solve non-linear recurrence relations (k_3)

UNIT 3: SET THEORY AND RELATIONS

Relations and ordering, Relations, Properties of binary Relations in a set, Relation Matrix and the Graph of a Relation, partition and covering of a set, Equivalence, Compatibility Relations, Composition of Binary Relations, Partial ordering, Hasse diagram, Principle of Inclusion-Exclusion, Pigeonhole Principle and its applications.

Learning Outcomes:

After completion of this module student will be able to

- draw Hasse Diagram for the given poset (k_3)
- apply principle of inclusion and exclusion and pigeonhole principle to real world problems (k_3)

UNIT 4: GRAPH THEORY

Basic Concepts, Representation of Graph, Sub graphs, Multigraphs, Planar graphs, Euler Paths, Euler circuits, Hamiltonian Graphs and Graph Isomorphism and its related Problems, Chromatic Number.

After completion of this unit student will be able to

- identify different graphs and their properties (k_3)
- construct Euler and Hamiltonian graphs (k_3)
- construct the graph for the given data (k_3)

UNIT 5: TREES

Spanning Trees, minimal Spanning Trees, BFS, DFS, Kruskal's Algorithm, Prim's Algorithm, Binary trees, Planar Graphs.

Learning Outcomes:

After completion of this unit student able to

- construct the spanning tree and binary tress from graphs (k_3)
- build minimal spanning tree by using different algorithms (k_3)

Text Books:

1. J.P. Tremblay and R. Manohar, Discrete Mathematical Structures with Applications to Computer Science, Tata McGraw Hill, 1997.
2. Joe L. Mott, Abraham Kandel and T. P. Baker, Discrete Mathematics for computer scientists & Mathematicians, 2/e, Prentice Hall of India Ltd, 2012.

References:

1. Keneth. H. Rosen, Discrete Mathematics and its Applications, 6/e, Tata McGraw-Hill, 2009.
2. Richard Johnsonburg, Discrete Mathematics, 7/e, Pearson Education, 2008
3. Narsingh Deo, Graph Theory with Applications to Engineering and Computer Science, Prentice Hall of India, 2006.